

50.002 Team 12

RhythMania

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Tagline

A beat matching rhythm game

Description

This project is an interactive rhythm game inspired by Rhythm Heaven designed to evaluate a player's musical timing, reaction speed and beat synchronisation. The system is housed in a laser-cut wooden enclosure that integrates input, audio and visual feedback components into a compact physical interface.

Gameplay begins when the player presses the button in the middle to initiate the game. A set of 4 LEDs emits a metronome signal at a defined 60 BPM (beats per minute). The player must press the button in precise alignment with each beat. The FPGA continuously measures the difference between the expected beat interval and the player's input, enabling real time accuracy evaluation.

Visual feedback is provided through the two 7 segment displays and the LED arrays around it. The 7 segment display tracks and updates the player's score dynamically based on timing precision.

Combining the numerical scoring system with the visual cues by the LEDs creates a feedback loop that allows players to calibrate their timing instinctively.

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